

CSS for Layout and Data Presentation

Box model • cascade • specificity
Flexbox • Grid • responsive • BEM

```
css/styles.css
```

Web Application Development • Semester 1, 2026–2027 • MSc. Tran Khai Minh • tkminh@hcmiu.edu.vn

What you already have

Structure

semantic HTML
h1 → h2 → h3

Navigation

multi-page
deep links, GET form

HTTP

request per resource
200 • 301 • 304 • 404

Lab 02 = presentation only. Your HTML stays exactly as Lab 01. No framework, no inline styles.

DevTools, Lab 01 — Network tab

DevTools, Lab 02 — Elements • Computed • device

Rule — structure first, presentation second

Same HTML, one new layer

LAB 01

Structure only

Semantic meaning

Document thinking

HTTP mechanics

DevTools Network

LAB 02

Structure + presentation

Visual hierarchy

Component thinking

Cascade • specificity

DevTools Elements + Computed

No Bootstrap • no Tailwind • no inline styles • no **!important**

Every element is a box

THE TRAP

```
/* content-box (default) */
.box {
  width: 200px;
  padding: 20px;
  border: 2px solid black;
  margin: 10px;
}

/* total = 200 + 40 + 4 + 20 = 264px */
```

THE FIX

```
/* what every project starts with */
* {
  box-sizing: border-box;
}

.box {
  width: 200px;
  /* total = 200px */
}
```

width: 200px sets the CONTENT width, not the element width. Use **box-sizing: border-box**.

How CSS finds elements, and which rule wins

SELECTOR TYPES

```
p           { } /* element  → 1 */
.intro      { } /* class   → 10 */
#header     { } /* id      → 100 */
nav a       { } /* descendant */
nav > ul > li { } /* child */
a:hover     { } /* pseudo-class */
tr:nth-child(even) { } /* zebra */
input[type="email"] { } /* attribute */
```

HIGHER SPECIFICITY WINS

```
p           { color: black; } /* 1 */
.intro      { color: blue;  } /* 10 wins */
#special    { color: red;   } /* 100 wins */
#header .nav a { } /* 111 wins */
```

Weight of each selector

Inline style – 1000

ID – 100

class • attribute – 10

element – 1

Practical rule

prefer classes over IDs

keep IDs for anchors only

one class per component, BEM

avoid chains like #header nav ul li

a long chain later forces !important

tie on specificity? the later rule wins

In DevTools: applied rules look normal, losing rules are struck out – hover one to see what beat it.

Origin first, then specificity, then order

1 • Origin

your styles beat browser defaults
(inline • style tag • external file)
between your own styles:
specificity, then order

2 • Specificity

inline 1000 • ID 100
class or attribute 10
element 1
compare the numbers

3 • Order in the file

```
p { color: red; }  
p { color: blue; }
```

the second one wins

Inheritance

inherits: color, font-family, text-align
does NOT inherit: border, padding,
margin, width, height

Origin → **specificity** → **order**. And check what inherits before you blame the cascade.

One dimension or two

FLEXBOX • 1 DIMENSION

```
.container {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  gap: 20px;  
}  
  
.item { flex: 1; } /* = 1 1 0 */
```

GRID • 2 DIMENSIONS

```
.page-layout {  
  display: grid;  
  grid-template-areas:  
    "header header"  
    "sidebar main"  
    "footer footer";  
  
  grid-template-columns:  
    200px 1fr;  
}
```

Flexbox = one row or column. Grid = rows and columns together. Layout decides with Grid; content decides with Flexbox.

Design small, then add

MOBILE-FIRST PATTERN

```
/* base = mobile 320+ */
.grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 1rem;
}

@media (min-width: 768px) {
  .grid { grid-template-columns: 1fr 1fr; }
}

@media (min-width: 1024px) {
  .grid { grid-template-columns: repeat(3, 1fr); }
}
```

Mobile — 320–767 • base, no query

Tablet — min-width: 768px

Desktop — min-width: 1024px

Large — min-width: 1440px

Test — Ctrl+Shift+M

Always try — 320 • 375 • 768 • 1024

Breakpoints follow the content, not the device. Test 320 too, that is where layouts break.

Make data comparable, forms usable

TABLE FOR SCANNING

```
table {
  width: 100%;
  border-collapse: collapse;
}
thead th {
  background: #f0f0f0;
  border-bottom: 2px solid #333;
}
tbody tr:nth-child(even) { background: #f9f9f9; }
tbody tr:hover { background: #e3f2fd; }
td.numeric {
  text-align: right;
  font-variant-numeric: tabular-nums;
}
.table-container { overflow-x: auto; }
```

FORM FOR USE

```
input, select, textarea {
  width: 100%;
  padding: 0.75rem;
  font-size: 16px; /* stops iOS zoom */
  border: 1px solid #ccc;
}
input:focus {
  outline: none; /* replaced, never removed */
  border-color: #06c;
  box-shadow: 0 0 3px rgba(0,102,204,.1);
}
input.error { border-color: #d32f2f; }
input:disabled {
  background: #f5f5f5;
  cursor: not-allowed;
}
```

Colour is information too: red for errors, green for success, grey for disabled. Check contrast on every one.

One value, many pages

CSS VARIABLES

```
:root {
  --color-primary: #06c;
  --color-text: #333;
  --spacing-md: 1rem;
}

.button { background: var(--color-primary); }

/* change here, every user updates */
[data-theme="dark"] {
  --color-primary: #64b5f6;
}
```

BEM NAMING

```
/* block */
.card { border: 1px solid #e0e0e0; }
/* element */
.card__title { font-size: 1.5rem; }
/* modifier */
.card--featured { border-color: #06c; }

<div class="card card--featured">
  <h3 class="card__title">Featured</h3>
</div>
```

File order: reset → variables → typography → layout → components → utilities → media queries

Elements, Computed, device toolbar

Open — F12 • Ctrl+Shift+I

Inspect — right-click → Inspect

Lost rules — struck out • hover for the winner

Computed — final pixel values

Box model — bottom of Styles panel

Responsive — Ctrl+Shift+M

Live editing

Change a value in the Styles panel
and the page updates instantly

but it is temporary. Copy the rule
into your css file or you lose it

Device toolbar: test 375, 768, 1024

Habit: finish a rule → device toolbar → check 375 and 1024 → then commit it.

IN CLASS • EXERCISE 1

Style your Lab 01 site

Base — your Lab 01 department site

Time — ≈ 60 minutes

Visual — fonts • sizes • line-height 1.5–1.8

Layout — header • nav • 1200px • footer

Nav — Flexbox • hover • current page

Responsive — 320px+ • nav vertical on mobile

File — css/styles.css

Max — 3 colours + shades

Technical requirements

external CSS file • no inline styles

CSS variables for colour + spacing

Flexbox navigation • 3 selector types

BEM naming • mobile-first queries

Explain in the report

colours + contrast check

Flexbox vs Grid decision

what changes mobile → desktop

DevTools: box model + override

Order of work: reset + variables → layout → typography → components → responsive last.

Same HTML, styled • 1200px

Welcome to the School

The School of Computer Science and Engineering is one of the schools of International University, Vietnam National University, Ho Chi Minh City. Its vision follows the vision of the University: to become a top research-oriented university of Vietnam with strong collaborations with world-wide prestigious universities, institutes, industries, local provinces and society.

Undergraduate programmes

Information Technology

7480201 • Bachelor of Engineering

Two specialisations: Network Engineering and Computer Engineering, covering

Computer Science

7480101 • Bachelor of Science

Majors in Computer Science and Software Engineering, with a strong industry project

Data Science

7480109 • Bachelor of Science

Managing, processing and analysing large data, with courses in data mining, machine

Your Lab 01 department site, styled • 1200px

One CSS file: reset + variables → typography → layout → components → responsive.

Same file at 390px



Same site at 390px • mobile

Base — mobile-first, min-width up

Phone tweaks — max-width: 767px

Nav — row → column

Cards — 3 across → 1

Table — padding 1rem → 0.5rem

Design for — 320px

Then check — 320 • 375 • 768

Shortcut — Ctrl+Shift+M

Check at 320 too — that is where most layouts break.

IN CLASS • EXERCISE 2

Style the course catalog and forms

Time — $\approx$ 75 minutes

Base — your Lab 01 course registration site

Home — hero • stats grid • call-to-action

Catalog — filter form • minimum 10 courses

Table — zebra • hover • sticky header

Numeric — right-aligned • tabular-nums

Sorting — visual indicators only

Form — grouped • focus • error • disabled

Confirm — success message • summary

Responsive — scroll • 1 column • stats 4 $\rightarrow$ 2 $\rightarrow$ 1

Explain in the report

table design + trade-offs

form usability + focus

responsive trade-offs

performance check

accessibility: contrast, keyboard

KEY RULES

```
thead th { position: sticky; top: 0; }  
.table-wrap { overflow-x: auto; }  
fieldset { border: 1px solid #ccc; }
```

Exercises 1 and 2 upload together: in-class item, before the session ends.

The multi-theme research site

Exercise 3 — ≈ 120 minutes + report

Base — your Lab 01 research lab site

Layout — Grid sidebar + results

Library — card • button • badge • alert

Tables — fixed header • expandable rows

Forms — multi-step progress • :checked fields

Themes — blue • green • dark

Page — one page showing every component

You decide, then justify

how to mark sortable columns
where error messages appear
wide table on mobile: scroll,
hide, or cards?
how a loading state should look

KEY RULES

```
/* dark mode = same variables, new values */
@media (prefers-color-scheme: dark) {
  :root { --bg: #1a1a1a; --text: #e0e0e0; }
}
```

Final item: Exercises 1–3 together, with the report.

The questions you answer in writing

Exercise 1

colours + contrast check
Flexbox vs Grid decision
mobile → desktop changes
DevTools: box model + override

Exercise 2

table design + trade-offs
form usability + focus
responsive trade-offs
performance check
accessibility: contrast, keyboard

Exercise 3 • 9 sections

architecture • themes • responsive
advanced CSS features
performance audit
accessibility • edge cases
real-world critique
+ a page showing every component

Show your work

screenshots of your pages
screenshots of the showcase page
DevTools with the box model open

COMMON MISTAKES

The ten ways this lab goes wrong

Inline styles → use a class, not style=

Heavy selectors → .nav-link--active

No border-box → * { box-sizing: border-box; }

Fixed px everywhere → max-width + media queries

Bad contrast → $\approx 4.5:1$ for body text

Focus removed → replace with a ring

!important everywhere → fix the selector instead

Hard-coded hex → var(--color-primary)

No mobile test → 375 • 768 • 1024

Never re-read code → graded on quality too

Every one of these is in the guide, section 6. Read it before you submit, the markers will look for exactly these.

SUBMISSION • TWO ITEMS

In class, then the final report

In class — Exercises 1–2, before the session ends

In-class ZIP — WEB-Lab02-<ID>-<Name>-InClass.zip

Final — Exercises 1–3 + report

Deadline — 23:59, date on the Blackboard item

Format — 1 ZIP ≤ 100 MB • PDF + DOCX

Final ZIP — WEB-Lab02-<ID>-<Name>.zip

Late — -30% • closed after 24 h

Attempts — last upload graded

Before you zip

ex1/ ex2/ ex3/ all in the Final ZIP
even what you uploaded in class
all pages load, no broken path
works at 320 / 768 / 1024
screenshots in the report
code typed, not screenshotted

Lab 01 final not accepted yet? Tell me before the end of the session — Exercise 1 is the same site.

WRAP-UP

Five things to remember

- 1 — box-sizing: border-box, always
- 2 — low specificity • one class • BEM
- 3 — cascade: origin → specificity → order
- 4 — Flexbox 1 axis • Grid 2 axes
- 5 — mobile first • test 375 / 1024

Next: Lab 03 — JavaScript, DOM and fetch

Questions: tkminh@hcmiu.edu.vn